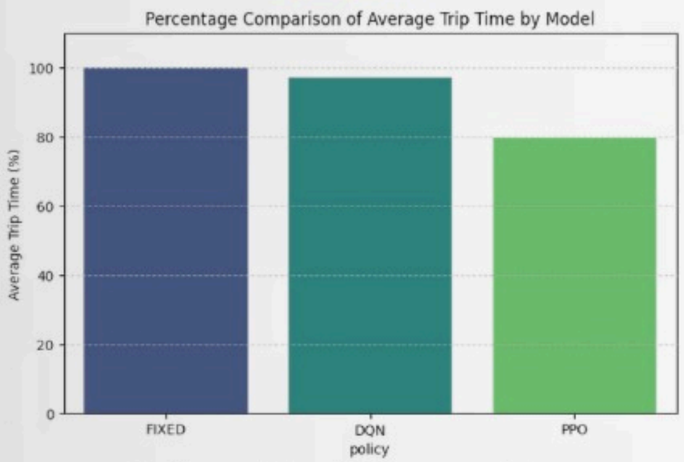
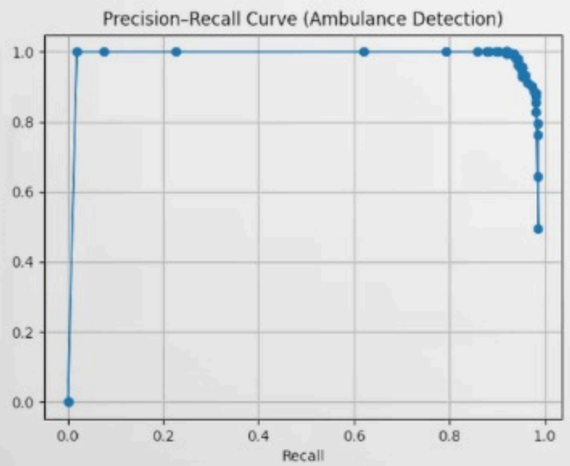
VISTA - VISION INTEGRATED SMART TRAFFIC AUTOMATION

Problem Statement

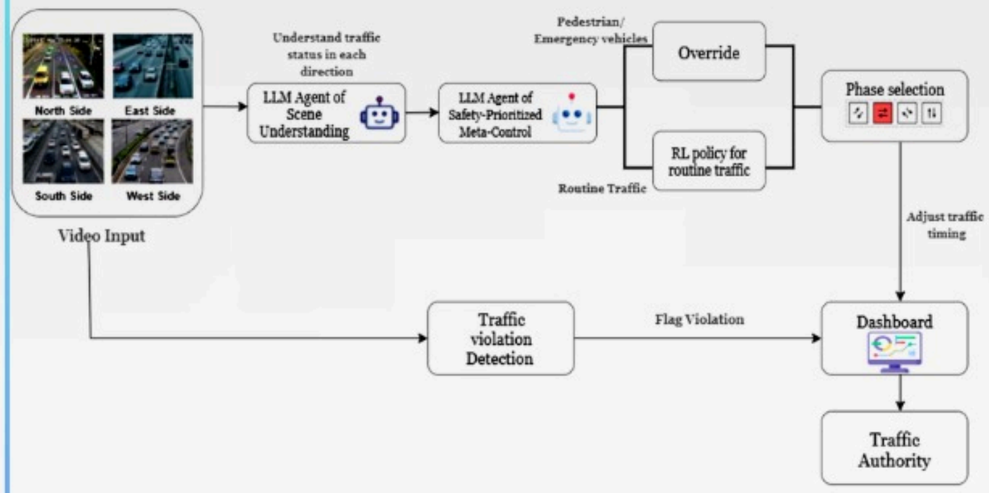
Indian cities face severe traffic congestion at intersections due to rapid urbanisation and a mixed traffic environment of cars, buses, two-wheelers, and pedestrians. Conventional fixed-time traffic signals are not adaptive to real-time conditions – they ignore traffic density, emergency vehicles, and pedestrian presence – resulting in inefficient flow, longer wait times, excess fuel consumption, and increased emissions.

Results

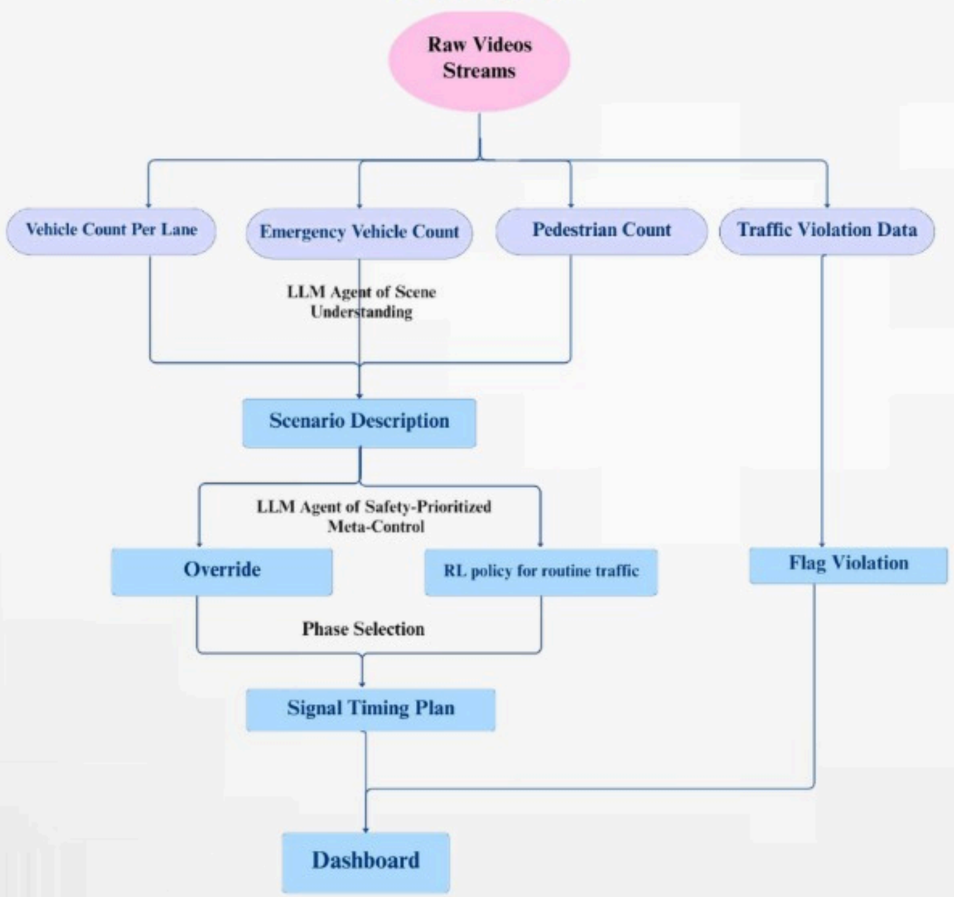




System Architecture



Workflow



Objective

- Dynamic traffic signal control using real-time detection and reinforcement learning
- Priority handling for emergency vehicles using AI-based recognition and control
- Improved pedestrian safety using YOLO-based detection
- Automated traffic violation detection and logging

References

[1] Emergency Response Optimization System with Dynamic Traffic Signal Control",ICoICI,2024.

[2] Dynamic Traffic Signal Optimization and Emergency Vehicle Prioritization",IDCIoT,2025.

[3] Traffic Optimization and Automated Emergency Vehicle Response System",International Conference on Recent Advances in Information Technology,2025.

[4] Smart Traffic Violation Detection System using Artificial Intelligence",ICPCSN,2023.

Conclusion

The conclusion summarizes the system's objective of improving urban traffic management using computer vision and reinforcement learning. It highlights its effectiveness in dynamic signal control and its potential for handling complex scenarios, providing a concise overview of its contribution to smarter and more efficient transportation systems.